

Identification of a bistable normal form for regime transitions in a driven dissipative transport system

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Abstract

Spatially extended dissipative systems under strong forcing can switch abruptly between distinct macroscopic transport regimes. We propose a hybrid reduced-order model in which a one-dimensional diffusion equation with a state-dependent transport coefficient is coupled to a scalar latent variable governed by the cusp normal form $\tau \dot{\zeta} = a(\mathbf{w}) + b\zeta - \zeta^3$. The two stable branches of the cusp define the two transport regimes; the transition and its hysteresis are saddle-node bifurcations of the latent dynamics, and the distance of the exogenous drive from the fold provides a quantitative, closed-form margin to transition. All model components — the transport coefficient, a residual source term, the drive map, and an observation map from the latent state to a measured emission proxy — are identified end-to-end from experimental time series by differentiating through an implicit–explicit integrator. Unlike proxy-following approaches, the latent dynamics are driven exclusively by exogenous actuation, so the inferred regime state is falsifiable against held-out observables. The identified model assigns a calibrated regime probability at every instant, locates transition times without supervision beyond weak segment labels, and yields bifurcation diagrams that summarize when a transition can be forced or undone.

1 Introduction

Abrupt transitions between metastable states are a recurring motif in nonequilibrium systems: bistability, hysteresis, and critical slowing down near folds are generic features predicted by normal-form theory (Kuznetsov, 2004; Strogatz, 2018). In the experimental system considered here, a driven dissipative transport process switches between a low-transport-barrier regime (*state L*) and a high-transport-barrier regime (*state H*) as the external drive increases, with the reverse switch occurring at a lower drive — the signature of hysteresis. A quantitative, model-based answer to the question “*which regime is the system in, and how far is it from switching?*” is a prerequisite for control.

Our contributions are: (i) a hybrid mechanistic–statistical model in which the regime is a latent coordinate of a cusp normal form, giving the L/H classification a precise mathematical meaning (basin membership of a bistable vector field); (ii) end-to-end identification through a differentiable implicit–explicit solver; and (iii) closed-form bifurcation diagnostics — fold margins $\mu_{\text{LH}}(t)$ and $\mu_{\text{HL}}(t)$ — that quantify proximity to forced and spontaneous transitions.

2 Model

2.1 Transport backbone

Let $u(\rho, t)$ denote the transported scalar field on the normalized domain $\rho \in [0, 1]$ with Neumann condition at $\rho = 0$ and a measured Dirichlet trace at $\rho = 1$. The field obeys a conservative

diffusion law with a learned residual source s_θ ,

$$\partial_t u = \frac{1}{V'(\rho)} \partial_\rho (V'(\rho) \chi(\rho, \zeta) \partial_\rho u) + s_\theta(\rho, u, n, \mathbf{w}, \zeta), \quad (1)$$

where V' is a fixed geometric weight and n an auxiliary measured field. The transport coefficient is a monotone boundary-layer profile whose edge value is modulated by the latent regime coordinate ζ ,

$$\chi(\rho, \zeta) = \chi_0 + \sigma\left(\frac{\rho - \rho_c}{w}\right) [\chi_1 - \Delta \chi \beta(\zeta) - \chi_0], \quad \beta(\zeta) = \sigma\left(\frac{3\zeta}{\sqrt{b}}\right), \quad (2)$$

so that the H regime ($\zeta > 0$) suppresses transport in the boundary layer and steepens the stationary profile there.

2.2 Latent regime dynamics: the cusp normal form

The regime coordinate evolves according to

$$\tau \dot{\zeta} = a(\mathbf{w}) + b\zeta - \zeta^3, \quad b > 0, \quad (3)$$

with drive $a(\mathbf{w}) = \boldsymbol{\kappa}^\top \mathbf{w} + \kappa_0$ an affine map of the exogenous inputs only (injected power, current, mean density). For $|a| < a_f = 2(b/3)^{3/2}$ the vector field (3) is bistable: stable equilibria $\zeta_L^* < 0 < \zeta_H^*$ are separated by a saddle ζ_s . Sweeping a through $+a_f$ annihilates the L branch in a saddle-node bifurcation (forced L→H transition); sweeping back through $-a_f$ annihilates the H branch (H→L back-transition). Hysteresis is therefore intrinsic rather than imposed. We define the *fold margins*

$$\mu_{LH}(t) = a_f - a(\mathbf{w}(t)), \quad \mu_{HL}(t) = a(\mathbf{w}(t)) + a_f, \quad (4)$$

which measure the distance of the instantaneous drive from the two folds and constitute the control-relevant output of the identified model.

2.3 Observation model and regime probability

The latent state is linked to a measured emission proxy $y(t)$ (bright in state L, quiescent in state H) through a learned map $\hat{y} = h_\psi(\beta(\zeta), \mathbf{w}, u_{\text{edge}}, n_{\text{edge}})$. Crucially, y never enters the latent dynamics (3); it only supervises the observation head, which breaks the circularity of proxy-following formulations. The instantaneous regime probability is the calibrated logistic

$$p_H(t) = \sigma\left(k \zeta(t) / \sqrt{b}\right), \quad (5)$$

and the mathematically sharp classification is basin membership: the system is in regime H at time t iff $\zeta(t)$ lies in the basin of attraction of $\zeta_H^*(a(t))$.

3 Identification

Data. Publicly released measurement campaigns from a toroidal confinement experiment (MAST); each discharge provides profile measurements $u(\rho_i, t_j)$ in the outer domain, exogenous inputs $\mathbf{w}(t)$, and the emission proxy $y(t)$ at millisecond resolution. Discharges are screened by strict per-channel quality gates; regime segment labels are produced by a dwell-time-constrained bimodal split of the lower envelope of y and used only as weak supervision.

Objective. The loss combines (i) a robust profile misfit, (ii) a source-magnitude penalty, (iii) smoothness and magnitude regularization of ζ , (iv) a binary cross-entropy on p_H against the weak segment labels, and (v) the observation-head misfit on y . Gradients are obtained by reverse-mode differentiation through a θ -method implicit–explicit integrator with tridiagonal implicit diffusion.

4 Results

5 Discussion

References

- Yuri A. Kuznetsov. *Elements of Applied Bifurcation Theory*. Springer, New York, 3 edition, 2004.
- Steven H. Strogatz. *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*. CRC Press, 2 edition, 2018.